Design a ladder that identifies the question you ask

This companion expands notes §§5 and 8.3–8.4, and blog §§5 and 8.3–8.4. Its focus is the relationship between the grid, statistical assumptions and engineering decision.

Choose the geometry before the solver

A fixed-TPP line uses $D = qN$. It can support forecasting along that recipe, but provides no independent intervention on size versus tokens. In a log-linear model, $\log D = \log q + \log N$ makes the two predictor columns collinear. More restrictive nonlinear forms can sometimes infer parameters along this line; that inference depends on the imposed form and does not replace observing off-line configurations.

For allocation, use multiple IsoFLOP slices or a two-dimensional size/token grid. For a slice, solve the actual compute function for each candidate horizon; $D = C/(6N)$ is only a starting approximation. If the apparent minimum is at the edge, expand the slice before fitting a frontier. Chinchilla, §3.

Concrete experiment: AI2's 16 runs

AI2 uses four sizes—190M, 370M, 760M and 1.3B—and four horizons $D = 20N \times \{1, 2, 5, 10\}$. This gives variation in both size and horizon rather than one TPP trajectory. It predicts 7B/13B target task performance under its recipe, with peak LR, batch and warmup extrapolated from a target anchor. It is not an exhaustive frontier search at every configuration. Bhagia et al., §2.1/Table 1.

Illustrative design: fit three sizes at several horizons; use a fourth size for development and a larger unseen tier for final validation. Exact sizes, seeds and token budgets must come from measured throughput, noise and the target decision. Development runs used to choose a fit are not untouched test runs.

Fit a declared objective

For endpoints (N_i, D_i, L_i), one possible objective is

$$J(\theta) = \sum_i w_i \, \rho_\delta\Big(\log \widehat{L}_\theta(N_i, D_i) - \log L_i\Big) \,,$$

where ρ_δ is Huber loss, w_i are declared weights and θ contains the surface parameters. Fitting log loss and fitting raw loss emphasize different discrepancies. Positivity constraints do not make every nonlinear scaling fit globally convex. Multi-start optimization reduces sensitivity to initialization; it does not certify a global optimum.

Freeze the transformation, bounds, floor treatment, exclusions and validation criterion. Report alternative solutions with similar objectives but different target predictions. Besiroglu et al., Chinchilla replication; (Mis)Fitting survey.

Respect dependence between observations

Intermediate checkpoints share initialization and data history; annealing branches share their trunk. Count prefix compute once, preserve the parent-child lineage, and bootstrap at the independent-run or shared-trunk-cluster level. A branch is a new endpoint but not an independent initial training history.

For each bootstrap replicate, refit the selected model and recompute the candidate gap or feasible allocation. Merely placing intervals on the exponents leaves the decision uncertainty unanswered.

Diagnose errors before adding compute

Pattern	Plausible explanation	Useful next experiment
Similar configuration varies across seeds	Training/evaluation noise	Independent repeats
Residual curvature across size and horizon	Functional-form or recipe mismatch	Off-axis runs and alternative form
Only long-horizon points fail	Schedule, repetition or stability	Matched-exposure long runs
Loss forecast passes, task forecast fails	Calibration/metric mismatch	Task-specific continuous metrics

These are diagnostic hypotheses, not automatic causal diagnoses. **Required artifact:** a frozen run table, development/test assignments, solver specification, residual plots and decision intervals.

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